

Introduction to Data Analytics

Code: CSIT 606

Module 1 Session 3

Dr. Rashmi Vashisth

Global Information Storage Capacity in optimally compressed bytes

2007 ANALOG

19 exabytes

- Paper, film, audiotape and vinyl: 6 %
- Analog videotapes (VHS, etc): 94 %
- Portable media, flash drives: 2 %
- Portable hard disks: 2.4 %
- CDs and minidisks: 6.8 %

DIGITAL

- Computer servers and mainframes: 8.9 %

- Digital tape: 11.8 %

- DVD/Blu-ray: 22.8 %

- PC hard disks: 44.5 %
123 billion gigabytes

- Others: < 1 % (incl. chip cards, memory cards, floppy disks, mobile phones, PDAs, cameras/camcorders, videogames)

DIGITAL
280 exabytes

1986
ANALOG
2.6 exabytes

DIGITAL
0.02 exabytes

1993

2000

ANALOG STORAGE

DIGITAL
STORAGE

2002:
"beginning
of the digital age"
50%

% digital:
1 %

3 %

25 %

94 %

Applications Of Big Data Finance

Banking:

- Since there is a massive amount of data that is gushing in from innumerable sources, banks need to find uncommon and unconventional ways in order to manage big data.
- It's also essential to examine customer requirements, render services according to their specifications, and reduce risks while sustaining regulatory compliance.

Stock Exchange:

- NYSE generates about one terabyte of new trade data every single day.
- So imagine, if one terabyte of data is generated every day, in a whole year how much data there would be to process.





- Social media in the current scenario is considered as the largest data generator.
- The stats have shown that around 500+ terabytes of new data get generated into the databases of social media every day, particularly in the case of Facebook.

Applications of Big Data: Healthcare

- Nowadays, doctors rely mostly on patients' clinical records, which means that a lot of data needs to be gathered, that too for different patients.
- Since there is a large amount of data coming from different sources, in various formats, the need to handle this large amount of data is increased



E-Commerce

- Maintaining customer relationships is the most important in the e-commerce industry. E-commerce websites have different marketing ideas to retail their merchandise to their customers, to manage transactions, and to implement better tactics of using innovative ideas with Big Data to improve businesses.

Applications of Big Data: Education

The education sector holds a lot of information with regard to curriculum, students, and faculty.

The information is analyzed to get insights that can enhance the operational adequacy of the educational organization.

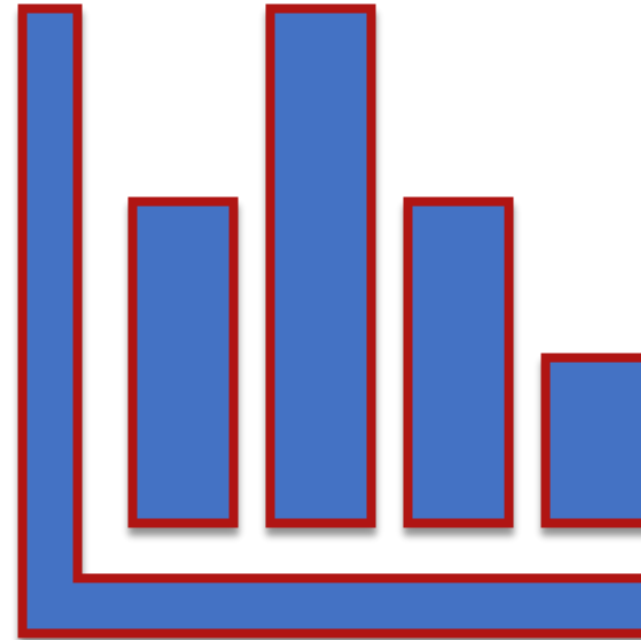
Collecting and analyzing information of a student such as attendance, test scores, grades, and other issues take up a lot of data.

So, big data makes an approach for a progressive framework wherein this data can be stored and analyzed making it easier for the institutes to work with.

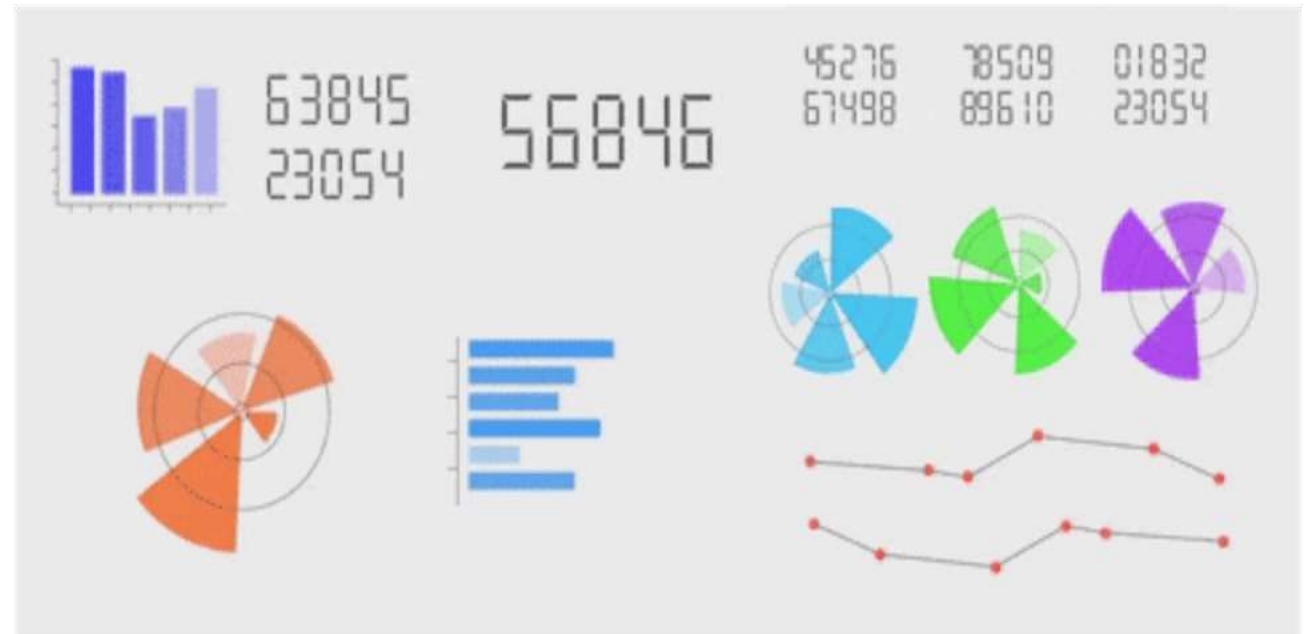
Analyzing limitations and solutions of existing data analytics

Objectives:

- Understanding Big data Analytics
- Difference between data analytics and Big data analytics
- Limitations
- Solutions



- Big Data Analytics examines large and different types of data in order to uncover the hidden patterns, insights, and correlations.
- Big Data Analytics is helping large companies facilitate their growth and development.
- It majorly includes applying various data mining algorithms on a certain dataset.



THANK YOU